

Siwar Derbeli Software Engineer - Specialization: Software Architecture

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🔗 <https://sywarder.github.io/portfolio-siwarderbeli/> 🐙 github.com/SiwarDerbeli 🇹🇳 Tunisian

🎓 EDUCATION

Engineering Degree in Computer Science, Higher Institute of Applied Science and Technology

2022 – 2025
Sousse, Tunisia

Preparatory Cycle, Higher Institute of Applied Science and Technology

2020 – 2022
Sousse, Tunisia

💼 PROFESSIONAL EXPERIENCE

Software Developer Intern (Erasmus+ Mobility), IES Cánovas del Castillo, Málaga, Spain - HTML, CSS, JS, PHP

04/2026 – Current

- Designed and executed a user-friendly layout for the school intranet.
- Engineered database indexing strategies, accelerating search and query performance for tables, reducing average data retrieval.

React Native Developer, Webgear - React Native, Websocket, Typescript

11/2025 – 03/2026
Freelance

- Developed a mobile operations app for hotel staff, supporting daily workflows within a management system
- Built 31 reusable components supporting 16 features, including 5 real-time functionalities, following atomic design principles to ensure a scalable and well-structured architecture

Software Engineer Intern, Enova Robotics - Spring Boot, Ionic React, Typescript, MySQL, InfluxDB, MQTT, Flask

02/2025 – 05/2025
Sousse, Tunisia

- Implemented a control and monitoring application for a solar panel cleaning robot, integrating InfluxDB and MQTT, reducing communication latency to ~50–100ms for real-time robot control.
- Extended the application with a **multi-robot simulation module**, supporting concurrent robot behavior (multithreading), GPS navigation, and real-time metrics generation.
- Integrated advanced features, including scheduling, automated task triggering, real-time alerts, remote control, and 3D robot behavior simulation with **ThreeJS** to enhance operational efficiency.
- Deployed a **deep learning-based anomaly detection model (89% accuracy)**, orchestrated with **Apache Airflow**, improving fault detection reliability.

SAP Consultant Intern, Deloitte Consulting - ABAP RESTful Programming, SAP UI5

06/2024 – 08/2024
Tunis, Tunisia

- Designed a complaints management web application on SAP BTP, supporting 2 key user roles (Employee, Quality Manager)..
- Delivered a fully integrated end-to-end SAP solution, aligning backend services with modern Fiori-based front-end design for an intuitive user experience.

Web Development Intern, Andromedia - React, Firestore, Redux

07/2023 – 08/2023
Tunis, Tunisia

- Developed an administrative dashboard for a digital comic book platform with efficient state management.

🛠️ TECHNICAL SKILLS

Software Development: HTML, CSS, JavaScript, Tailwind, MUI, Ant Design, React JS, Ionic React, React Native, Python, Flask, FastAPI, Node.js, Express.js, Spring Boot, ABAP RESTful

Data & Databases: Databases: MongoDB, PostgreSQL, MySQL, InfluxDB | Data Science & ML: Pandas, NumPy, Scikit-learn, TensorFlow

Cloud, DevOps & Streaming: Tools & DevOps: Git, GitHub, Docker, CI/CD (GitHub Actions) | Cloud & AI: Azure AI & Machine Learning, Cognitive Services | Streaming & Pipelines: MQTT, Kafka, Telegraf, Apache Airflow

📁 PROJECTS

Academic Curriculum Tracker, ReactJS, Spring Boot, Tailwind, MySQL

- Defined **user management, scheduling, and note tracking features**, improving task organization and saving administrators an estimated **20% of schedule planning time**
- Followed a **hexagonal architecture** for maintainability and scalability, and delivered an **intuitive, user-friendly interface**.

Messaging App, MERN stack, Socket.io

- Engineered a **Messaging Application** using the **MERN stack** and **Socket.io** 🔄, enabling **real-time, bidirectional communication** via WebSockets.
- Created a responsive UI with reusable components, **enabling faster feature integration and consistent design**

University Management System, Spring Boot, Kafka, Docker, AOP

- Managed **student and teacher information** through a **decentralized system**, implementing the **CQRS model** with **Apache Kafka** for asynchronous microservice communication.
- Created AOP-based logging and monitoring, standardizing error tracking across the application and accelerating issue resolution

AI-Powered GitHub Documentation Generator, React, Tailwind, FastAPI

- Built a web app that generates **documentation** from **GitHub repositories** using **GitHub API** and **GitPython**
- Applied **AI summarization** to produce structured, readable documentation.

 LANGUAGES

French



English
7.5 in IELTS



 Certificates

- Data Science Bootcamp:** 365 Careers
- Learn React:** Scrimba
- Prepare for Your Career in Generative AI:** LinkedIn & Microsoft
- Foundations of Artificial Intelligence:** LinkedIn & Microsoft
- GenAI Job Simulation:** Forage
- Azure AI Fundamentals:** Microsoft

 Organizations

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|---|-------------|
| CyberTrace Club, <i>Marketing Committee Member</i> | 2022 – 2023 |
| CyberTrace Club, <i>Marketing Team Leader</i> | 2023 – 2024 |